Cycle Self-Training With Joint Adversarial for Cross-Scene Hyperspectral Image Classification

Zhongwei Li, Yajie Yang, Leiquan Wang, *Member, IEEE*, Mingming Xu, *Member, IEEE*, Ziqi Xin, Jie Wei, and Yuewen Wang

***Abstract*— Cross-scene hyperspectral image classification (HSIC) leverages existing knowledge to categorize unknown scenes, aligning with the practical applications of remote sensing monitoring. However, spectral shifts across domains pose substantial challenges for this classification, aggravated by the insufficient number of labeled samples. Most existing methods predominantly address domain alignment from a singular perspective, rendering them inadequate to sustain robust classification performance in the presence of significant domain shifts. Additionally, although self-training can mitigate the labeling deficiency by leveraging unlabeled data, existing methods often fail to ensure the effective and accurate utilization of such data. Consequently, this article proposes a hyperspectral image (HSI) cross-scene classification architecture based on cycle self-training with joint adversarial (CSJA), which mitigates the impact of spectral shifts on cross-scene classification. Specifically, the proposed approach incorporates domain adversarial modules to reconcile domain distributions at varying granularities, coupled with a class adversarial module for joint adversarial alignment. Moreover, the cycle self-training (CST) module is devised to explicitly enforce pseudo-label generalization, thereby fully harnessing the informative content of the target domain. To effectively exploit both spatial and spectral information in HSIs and extract discriminative features, a convolutionally enhanced Transformer feature extraction network is proposed to generate feature-rich representations for both domains. Experimental evaluations on two publicly available cross-scene hyperspectral datasets and two self-made UAV hyperspectral datasets validate the superiority of the proposed algorithm.**

***Index Terms*— Cross-scene classification, cycle self-training (CST), hyperspectral image (HSI), joint adversarial.**

I. Introduction

HYPERSPECTRAL images (HSIs) are remote sensing images with dozens or even hundreds of continuous and segmented spectral bands, containing rich spectral and spatial information [1]. By virtue of its advantage of spectral integration, HSIs are crucial in fields like wetland classification [2], water quality monitoring [3], and ecological quality monitoring [4].

HSI classification (HSIC) represents one of the most critical applications of HSI. In the initial phases of HSIC research, various traditional machine-learning methods were used [5], [6], [7], [8]. More recently, deep learning techniques have driven significant advances in HSIC [9], [10]. Deep learning techniques, utilizing network architectures like convolutional neural networks (CNNs) [11] and generative adversarial networks (GANs) [12], have recently emerged and gained prominence. The effectiveness of HSIC depends heavily on feature extraction, and methods like the feature complementary attention network (FCAN_AKF) [13] have been developed to improve this aspect by capturing nonlocal spectral features. Another challenge in HSIC is the scarcity of labeled samples, which has been addressed using few-shot learning frameworks and positive feedback mechanisms [14], [15]. Although HSIC has achieved commendable results, the success of these methods typically requires a certain amount of labeled samples, which poses a significant challenge for HSIs [16], [17]. Furthermore, HSIC often focuses on single-scene classification, where both training and testing are conducted on the same HSI. However, practical applications often require learning from labeled HSIs to classify other images with similar land cover types, known as HSI cross-scene classification (HSICC) [18]. HSICC faces considerable challenges due to spectral shifts caused by varying acquisition conditions across different scenes, making it difficult for HSIC methods to achieve desired results [19]. Recently, transfer learning, especially unsupervised domain adaptation (UDA), has been employed to address these spectral shifts in HSICC. In UDA, knowledge is transferred from a labeled source domain to an unlabeled target domain to improve classification accuracy, with methods categorized into traditional manual designs and deep learning approaches [20].

Traditional methods in HSICC, primarily instance-based and classifier-based, rely on manual design. Instance-based methods focus on aligning sample marginal distributions between the source and target domains by adjusting sample weights during model training [21], [22], [23]. However, they often overlook spectral shifts in HSIs, thus limiting their effectiveness in HSICC. Classifier-based methods, on the other hand, adapt classifiers to domain differences using unlabeled target domain samples [24], [25]. Yet, these manual

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Zhongwei Li, Mingming Xu, and Ziqi Xin are with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: li.zhongwei@vip.163.com).

Yajie Yang, Leiquan Wang, Jie Wei, and Yuewen Wang are with the College of Computer Science and Technology, China University of Petroleum (East China), Qingdao 266580, China (e-mail: yyj_edu@163.com; richiewlq@gmail.com).

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approaches typically adjust at the sample and model levels only, struggling to achieve robust generalization due to feature diversity, resulting in unstable classification performance.

Deep learning methods are becoming increasingly important in domain adaptation (DA) due to their ability to learn high-level, abstract feature representations, which allows for the capture of complex nonlinear features in HSIs. These techniques refine feature representations to increase similarity between the source and target domains in the representation space, thus improving model alignment with the data. The two main frameworks that have emerged in this area are discrepancy-based methods and adversarial-based methods.

Discrepancy-based approaches focus on measuring the difference in features between the source and target domains and minimizing the distributional differences between them through various means. Maximum mean discrepancy (MMD) is a commonly used distance metric to measure the difference between two distributions, and many researchers have proposed a series of methods based on MMD and its variants [26], [27]. Long et al. [28] proposed an end-to-end deep DA method (DAN), employing the multiple kernel variant of MMD (MK-MMD) to align the feature distributions between the source and target domains. Building on DAN, they later proposed a joint adaptation network (JAN) [29], utilizing Joint MMD (JMMD) to adjust multiple domain-specific layers. Zhang et al. [30] proposed a transfer network that aligns distributions based on MMD and graph optimal transport (GOT). Although such methods are beneficial for precise feature selection and alignment, they may struggle to fully capture the complexities of diverse scene variations when relying on single-feature or simplistic difference analyses. Therefore, adversarial-based methods may achieve more stable results in complex scenarios.

In contrast to discrepancy-based methods, adversarial-based approaches, inspired by GANs, emphasize adversarial training to better align the distributions between the source and target domains. Domain adversarial neural networks (DANNs) [31] use gradient reversal layers for domain alignment during backpropagation, though this can lead to a loss of class-specific information [32]. Saito et al. [33] proposed MCD, an adversarial learning method that enhances classification by focusing on class-specific features, inspiring further research [34]. Building upon MCD, Huang et al. [35] proposed a two-branch attentional adversarial data processing network (TAADA) to effectively utilize HSI information. Class-adversarial methods enhance feature discrimination among classes [36], but significant domain shifts can still cause fluctuations in classification performance. Tang et al. [37] combined domain and class adversarial techniques, though their simplified domain adversarial process may lose essential domain-transferable features. Despite progress, substantial domain shifts highlight the need for more robust approaches.

Given the limited availability of labeled samples in HSIs, several researchers have employed self-training techniques for pseudo-label generation to leverage unlabeled target domain data [38], [39]. Wang et al. [40] proposed the bi-classifier adversarial augmentation network (BCAN) to identify reliable samples for inclusion in the training set using a thresholding approach, though its simplicity limits its reliability. Moreover, methods like confident learning [41], [42] aim to enhance the discriminative capability of neural networks by incorporating high-confidence samples. However, without supervision from target domain labels, pseudo-labels can be inaccurate, potentially harming network performance. Therefore, we focus on reducing the impact of erroneous information on the network's discriminative ability and strive to introduce more useful information from the source domain into the target domain.

To tackle these challenges, this article introduces a novel UDA architecture, cycle self-training with joint adversarial (CSJA), specifically designed to address domain shifts and the scarcity of labeled samples. CSJA enhances robustness against domain shifts by implementing joint adversarial techniques that simultaneously align domain and class distributions, ensuring a more stable adversarial effect. Additionally, the architecture incorporates cycle self-training (CST) method to fully utilize the unlabeled target domain data to expand the training, effectively expanding the training set and mitigating the limitations posed by the limited availability of labeled samples. These combined strategies significantly improves the transferability and discriminative power of features across domains. CSJA incorporates several key components. First, a novel local domain discriminator is designed to evaluate transferability information in tokens of the patch, which are then integrated into the Transformer's self-attention (SA) mechanism, guiding the network to focus on more transferable tokens. Furthermore, CSJA introduces a principled self-training algorithm called CST, which is designed to transfer useful information from source HSIs to target HSIs. It iterates between forward and backward steps until convergence, enabling the generalization of pseudo-labels across domains. By avoiding traditional self-training methods, CSJA enhances the robustness of source data classification, even in the presence of spectral shifts. In comparison with recent advancements in this domain, the primary contributions of our proposed CSJA architecture can be summarized as follows.

1) We propose the CSJA architecture for the HSICC task, incorporating the unique data characteristics of HSIs to address the interdomain gap from multiple perspectives. It effectively preserves the transferability and discriminative features between the source and target domains.
2) We develop a feature extraction module based on the Transformer architecture and designed a dimensional convolution to extract high-dimensional features from HSIs. By integrating local spatial features with global contextual features, our method achieves a comprehensive representation of spatial–spectral information.
3) We devise a global–local domain discriminator for multigranularity domain alignment. The global discriminator ensures coarse alignment, while the local discriminator integrates transferability into the Transformer's SA, enhancing alignment for local features and directing the feature extraction network to focus on transferable and discriminative features for precise alignment.
4) We implement a CST module specifically designed for knowledge transfer in cross-scene HSIs to overcome the limitations of standard self-training. This module iterates

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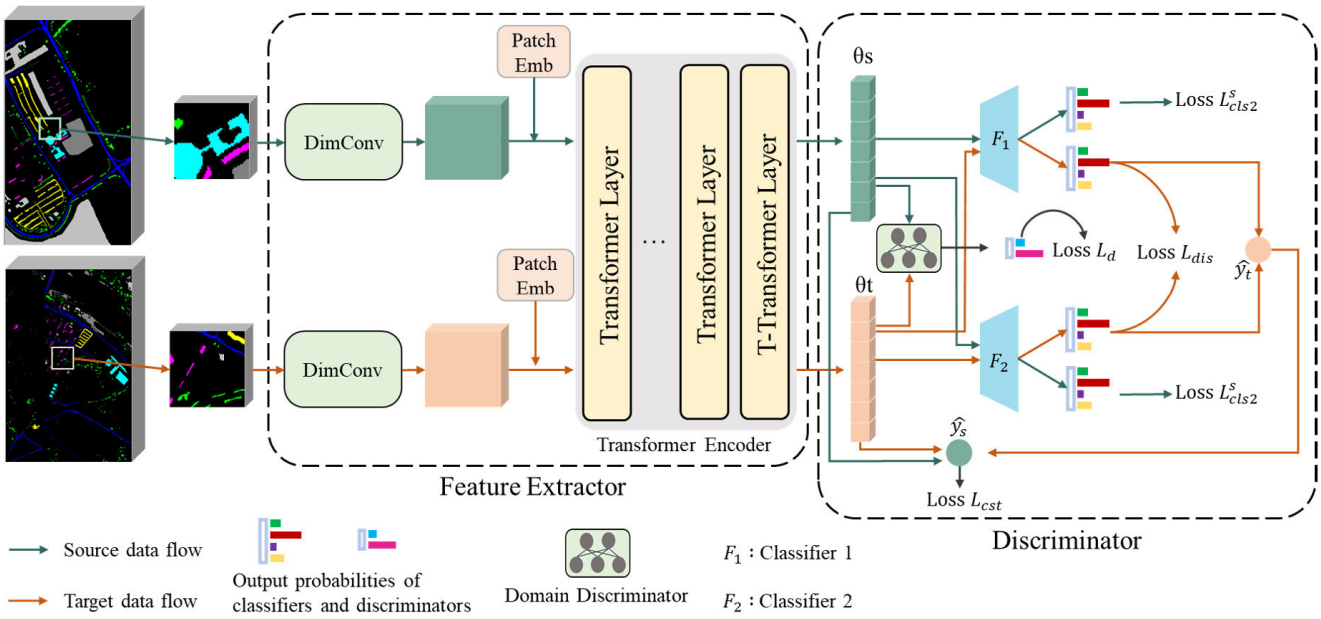


Fig. 1. Overall architecture of CSJA.

between training the classifier with target pseudo-labels and updating the shared representation, ensuring robust classifier performance on source data and explicitly enforcing pseudo-label generalization.

The article is organized as follows. Section II presents a comprehensive overview of the CSJA architecture. Section III delves into the experimental findings, while Section IV provides a summary of the article.

II. Proposed Method

A. Overall Network Architecture

The architecture of CSJA, as depicted in Fig. 1, comprises two components: the feature extractor and the discriminator. The feature extractor includes a convolutionally enhanced Transformer network, while the discriminator consists of two domain discriminators (one embedded in the T-Transformer layer of the feature extraction network) and two classifiers. Each domain discriminator comprises m layers of FC-ReLU-Dropout stacks, and each classifier adopts an n-layer FC-BN-ReLU-Dropout configuration.

Specifically, HSIs are input as square patches, centered around the pixel of interest, and subjected to feature extraction by a CNN-enhanced Transformer network to construct high-level feature representations. The extracted features from both the source and target domains are then separately utilized in the classifier and domain discriminator for adversarial training and CST, aiming to mitigate the domain gap. These two modules alternate iteratively to enhance classification accuracy. During testing, the combined predictions of the two classifiers are employed as the test results for the target domain.

B. Transformer-Based Feature Extractor Enhanced With Convolution

In recent years, vision transformer (ViT) has been on the rise [43]. Yang et al. [44] conducted a comprehensive analysis and experimental evaluation of the effectiveness of the ViT in DA tasks. The study results suggested that ViT outperformed robust methods based on CNNs, even without any additional adjustments. In addition, patch tokens in ViT capture local features across various positions, while the SA mechanism effectively extracts critical information. These attributes confer upon ViT's substantial potential for DA. We fully leverage the potential of Transformer networks for data processing and have designed a dimensional convolution module tailored to the characteristics of HSIs. This structure extracts high-dimensional local features from HSIs, complementing the global contextual features extracted by the Transformer. This design comprehensively expresses the spatial–spectral information of HSIs, facilitating further alignment tasks.

The feature extractor (denoted as E) aims to comprehensively extract rich feature representations from the input data in preparation for subsequent processing. It is well known that ViT can establish long-range dependencies in sequence data to capture global features, particularly suited for extracting high-dimensional spectral information features in HSIs. Convolution excels in extracting local features and capturing spatial features not addressed by ViT. The feature extractor of CSJA integrates convolutional enhancement into the backbone network of the Transformer model, preserving remote contextual relationships while also incorporating local attributes. Assuming the labeled source samples HSI are represented as $\mathbf{X}^s = \{\mathbf{x}_i^s\}_i^{n_s}$, with their corresponding label set $\mathbf{Y}^s = \{y_i^s\}_i^{n_s}$. The unlabeled target samples are represented as $\mathbf{X}^t = \{\mathbf{x}_i^t\}_i^{n_t}$. n_s and n_t are the number of samples. It is worth noting that the labels of the target domain are not used in the training.

We design a dimensional convolution module (denoted as DimConv) to enhance the feature representation capability of ViT. DimConv not only extracts local features like regular convolution but also takes into account the information contained in the channel dimension, making it suitable for the intricate dimensions present in HSIs. First, patches of HSIs are

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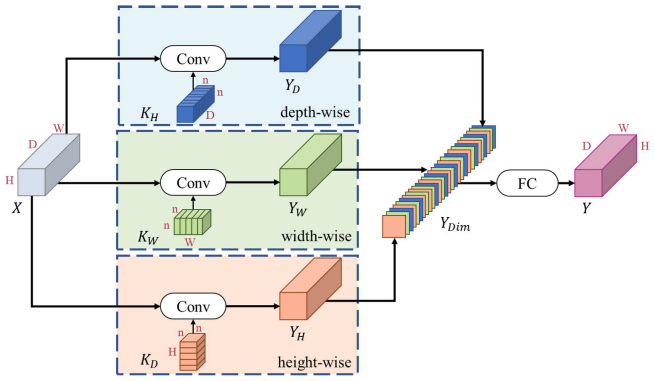


Fig. 2. Encoding procedure of DimConv for the three dimensions of the input tensor.

independently encoded over the depth, height, and width using DimConv to learn local dimensional representations. As shown in Fig. 2, DimConv encodes each dimension of each patch as a branch for processing. Each of these three branches employs a distinct convolution kernel along various directions, followed by interleaving the convolved features in the depth direction to produce the output $\mathbf{Y}_{\text{Dim}} \in \mathbb{R}^{3D\times H\times W}$. Finally, to minimize computational complexity, we employ a fully connected (FC) layer to condense the channel representations, resulting in the output of DimConv as $\mathbf{Y} \in \mathbb{R}^{D\times H\times W}$.

Subsequently, the results $\mathbf{Y}$ obtained from DimConv are fed into the ViT model. Each $\mathbf{Y}$ corresponding to a patch is regarded as an input, and then the Transformer network with L Transformer blocks is used to model the global context of the input. Assuming that ψ represents ViT, the class token that summarizes all relevant information can be expressed as follows:

$$\mathbf{f}_i^* = \psi(\mathbf{Y}_i^*), \quad \mathbf{f}_i^* \epsilon \mathbb{R}^{1\times 1\times \hat{d}} \tag{1}$$

where $*$ denotes either the source (s) or target (t) domain, and $\mathbf{Y}_i$ represents the result of the ith patch after being processed by DimConv.

C. Global–Local Domain Discriminator

Domain adversarial training has been demonstrated to effectively address HSICC challenges. CSJA has designed two domain discriminators, the global and the local domain discriminators, which implement domain alignment at different granularities. The local domain discriminator promotes the network to prioritize features that contain transferable semantics and pay less attention to irrelevant features. Meanwhile, the global domain discriminator enforces alignment between the large-scale marginal distributions of domains. These two domain discriminators complement each other to achieve better domain alignment.

Considering that we use ViT as the feature extractor, which has attention mechanisms and sequential tokens, we propose embedding the local domain discriminator into the attention mechanism of Transformer block. Different tokens divided by patches represent localized information from various regions within the patch, offering a nuanced patch representation. Given tokens of varying semantic importance and transferability, the local domain discriminator can assign distinct weights to encourage class tokens, i.e., advanced feature representations amalgamating all learned output information, to pay more attention to transferability and discriminative tokens. While SA weights in ViT are often deemed discriminative, they may not explicitly reflect the transferability information of each token. To tackle these challenges, we employ the local domain discriminator D_l to achieve fine-grained cross-domain alignment

$$L_{D\text{-local}}(\mathbf{x}_i^{s,t}, d_i^{s,t}) = -\frac{1}{nR}\sum_{\mathbf{x}_i\in D}\sum_{r=1}^{R} L_{\text{ce}}\left(D_l\left(E_f(\mathbf{x}_{ir}^*)\right), d_{ir}^*\right) \tag{2}$$

where $\mathbf{x}^{s,t}$ and $d^{s,t}$, respectively, denote the data in the source and target domains and their corresponding domain labels, n is the sum of samples, R is the number of tokens in the patch, D is the ensemble of the source and target domains, L_{ce} stands for cross-entropy loss, E_f is the feature extractor, D_l is the local domain discriminator, $\mathbf{x}_{ir}^*$ represents the rth token of the ith patch, d_{ir}^* is the true domain label of $\mathbf{x}_{ir}^*$, and $D_l(\mathbf{f}_{ir})$ shows the probability that $\mathbf{f}_{ir}$ belongs to two domains using D_l, and $\mathbf{f}_{ir}$ denotes the data processed by the feature extractor. During training, D_l attempts to accurately discriminate the domain to which each token belongs, while E hopes to generate tokens that can confuse D_l so that D_l cannot accurately discriminate them. In theory, a token that is less easily recognized by D_l typically has greater transferability and should be given more attention. Therefore, we use $t_{ir} = T(\mathbf{f}_{ir}) \in [0, 1]$ to assess the transferability of the rth token of the ith patch.

Specifically, we embed the D_l into the multihead SA (MSA) of ViT, and modify the MSA into Transferable MSA (TMSA), which allows it to add the transferability of learned tokens to the attention weights. SA is the basis of MSA and it can be represented as

$$\text{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d}}\right)\mathbf{V} \tag{3}$$

$$\text{MSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\mathbf{head}_1, \ldots, \mathbf{head}_k)\mathbf{W}^{\mathbf{O}}$$
$$\text{where} \cdot \mathbf{head}_i = SA\left(\mathbf{Q}\mathbf{W}_i^{\mathbf{Q}}, \mathbf{K}\mathbf{W}_i^{\mathbf{K}}, \mathbf{V}\mathbf{W}_i^{\mathbf{V}}\right) \tag{4}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent query, key, and value vectors, respectively, d represents the dimension of $\mathbf{Q}$ and $\mathbf{K}$, $\mathbf{W}_i^{\mathbf{Q}}$, $\mathbf{W}_i^{\mathbf{K}}$, $\mathbf{W}_i^{\mathbf{V}}$, and $\mathbf{W}^{\mathbf{O}}$ are the weight matrices, Concat represents concatenating data, and $\mathbf{head}_1$ to $\mathbf{head}_i$ denote the output of i heads of the multihead subattention. Transferable SA (TSA) is described as

$$\text{TSA}(\mathbf{q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{q}\mathbf{K}^{\text{T}}}{\sqrt{d}}\right) \odot \text{T}(K_{\text{patch}})\mathbf{V} \tag{5}$$

where $\mathbf{q}$ represents the query of the class token, K_{patch} is the key of tokens, and $\odot$ is element-wise product. $\text{T}(K_{\text{patch}})$ is the probability that each patch can be correctly judged by D_l, and tokens that are not easy to be judged usually have higher transferability. This formula illustrates the semantic importance of each token with $\text{softmax}(\mathbf{q}\mathbf{K}^{\text{T}}/\sqrt{d})$, while $T(K_{\text{patch}})$ represents the transferability of individual

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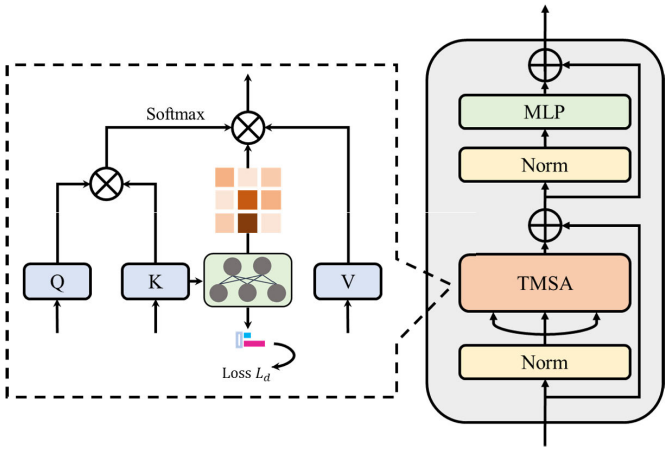


Fig. 3. Specific structure of the T-Transformer layer.

tokens. To handle complex semantic relationships, TMSA is formulated as follows:

$$\begin{aligned}\mathrm{TMSA}(\mathbf{q},\mathbf{K},\mathbf{V}) &= \mathrm{Concat}(\mathbf{head}_1,\ldots,\mathbf{head}_k)\mathbf{W}^{\mathbf{O}}\\ \text{where}\cdot\,\mathbf{head}_i &= \mathrm{TSA}\left(\mathbf{Q}\mathbf{W}_i^{\mathbf{Q}},\mathbf{K}\mathbf{W}_i^{\mathbf{K}},\mathbf{V}\mathbf{W}_i^{\mathbf{V}}\right).\end{aligned} \tag{6}$$

The transformer block, which substitutes MSA with TMSA, is termed the T-transformer layer. To prevent an overwhelming impact of TMSA during feature extraction, which could result in incomplete feature extraction, the T-transformer layer is exclusively employed in the final layer of the Transformer encoder. Thus, the transformer encoder consists of a total of L layers, where the initial $L-1$ layers are standard transformer layers, and the Lth layer constitutes the T-transformer layer, as depicted in Fig. 3. $L-1$ is set to 7 in this article.

As shown in Fig. 1, the global domain discriminator, denoted by D_g, receives the output class token from the feature extractor and identifies which domain the class token belongs to. The objective of D_g is to prevent D_l from compelling localized feature alignment between the two domains to disregard the overall distribution of both domains. Its corresponding loss function is denoted as follows:

$$L_{D\text{-global}}\left(\mathbf{x}_i^{s,t},d_i^{s,t}\right) = -\frac{1}{n}\sum_{\mathbf{x}_i\in D} L_{\mathrm{ce}}\left(D_g\left(E_f(\mathbf{x}_i^*)\right),d_i^*\right). \tag{7}$$

The symbols in (7) have the same meaning as in (2). D_g learns global domain-invariant features and complements D_l by globally reducing the disparity between the two domains at a coarse granularity.

D. Cycle Self-Training

Given the limitations of traditional self-training, we design CST to improve the poor performance of pseudo-labels under spectral shifts. CST iterates between forward and backward steps, allowing the model to excel in both source and target domains. The forward step is similar to standard self-training. First, the network is initially trained using source domain data to improve its discriminative ability. Subsequently, the results of the classifier are utilized to generate preliminary pseudo-labels for each sample in the target domain dataset

$$\widehat{y_i^t} = \max\left\{F_1\left(E_f(\mathbf{x}_i^t)\right)+F_2\left(E_f(\mathbf{x}_i^t)\right)\right\} \tag{8}$$

where F_1 and F_2 represent two classifiers. The final outcome is determined by combining the results of the two categorizations, and the category with the greatest likelihood $\widehat{y_i^t}$ is chosen as the pseudo-label for $\mathbf{x}_i^t$.

However, the accuracy of unsupervised pseudo-label prediction in deep networks is often limited [45], and it may not necessarily correlate with true confidence even in the absence of a distribution shift. Especially in the HSICC context, the spectral shifts between the source and target domains make the selection of pseudo-labels need to be more careful, and the commonly used pseudo-label selection strategies cannot guarantee the correctness of pseudo-labels. The purpose of CST is to discard simple threshold selection or reweighting, driving the model to gradually improve the pseudo-labels by itself and achieve effective classification in the target domain.

The reverse step is central to CST's ability to overcome the limitations of self-training. Information about the source domain contains both transferable information that can be used by the target domain and task-irrelevant information that is detrimental to feature matching, and using as much of the domain-transferable information as possible is the key to effectively classifying the target domain. This means that labels in the source domain contain both useful information that can be transferred to the target domain and harmful information that can lead to pseudo-labeling errors in the target domain. The performance of the model transferred to the source domain can reflect the quality of the pseudo-labels. By training the network directionally, we may encourage the pseudo-labels to contain more information useful for classification in the source domain. Therefore, we use target domain pseudo-labels and shared representations to predict samples from source domain, and jointly enforce classification to improve the network. Pseudo-label generation and CST are denoted as

$$\widehat{y_i^s} = C_{\widehat{y_i^t},\phi}(E(\mathbf{x}_i^s)) \tag{9}$$

$$L_{\mathrm{Cycle}} = \frac{1}{n_s}\sum_{\mathbf{x}_i^s\in D_s} L_{\mathrm{ce}}\left(\widehat{y_i^s},y_i^s\right) \tag{10}$$

where $C_{\widehat{y_i^t},\phi}(\cdot)$ represents a prediction of $(\cdot)$ based on $\widehat{y_i^t}$ and ϕ, ϕ denotes the shared representation on the source domain, y_i^s denotes the true label of $\mathbf{x}_i^s$. By employing pseudo-labels and shared representations to predict the source domain samples, the proximity of the predictions to the true labels corresponds to the convergence of the feature spaces projected from the two domains. It is suggested that the pseudo-labels predicted by the model contain more transferable information.

E. Implementation Details

We employ adversarial learning involving the feature extractor E, two classifiers F_1 and F_2, and domain discriminators D_l and D_g to minimize the impact of domain shift on the cross-scene classification. Specifically, we update E by minimizing the classifier discrepancy and maximizing the domain discrimination loss, optimize the dual classifiers by maximizing the

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classifier discrepancy, and update the domain discriminators by minimizing the domain discrimination loss. This adversarial learning considers both interdomain alignment and interclass alignment, which ultimately serves the purpose of aligning the distributions of the source and target domains. The total loss of CSJA can be expressed as

$$L(\mathbf{X}_s, \mathbf{Y}_s, \mathbf{X}_t) = L_{\text{cls}}(\mathbf{X}_s, \mathbf{Y}_s) + \gamma L_{\text{dis}}(\mathbf{X}_t) + \alpha L_d(\mathbf{X}_s, \mathbf{Y}_s) + \beta L_{\text{Cycle}}(\mathbf{X}_s, \mathbf{Y}_s, \mathbf{X}_t, \widehat{\mathbf{Y}_t}). \tag{11}$$

The overall loss comprises four main components: supervised classification loss for labeled source domain samples, discrepancy loss for unlabeled target domain samples across both classifiers, domain discrimination loss, and CST loss. The symbols γ, α and β are weighting parameters for the mixed loss. In CSJA, individual modules are trained using these losses in four steps.

In Step 1, to pretrain the network, we minimize the classification loss of the source domain to optimize the parameters θ_g, θ_{f_1}, and θ_{f_2} of the feature extractor E and classifiers F_1 and F_2, respectively, so that the features possess initial discriminability

$$L_{\text{cls}}(\mathbf{x}_i^s, y_i^s) = \frac{1}{2n_s} \sum_{i=1}^{n_s} \sum_{j=1}^{2} L_{\text{ce}}(F_j(E_f(\mathbf{x}_i^s)), y_i^s) \tag{12}$$

where $F(\cdot)$ is the joint prediction of F_1 and F_2. Additionally, this step encourages E to generate more similar features in the source and target domains by maximizing the domain discrimination loss, thus confusing the domain discriminators. The domain discrimination loss can be expressed as

$$L_D\left(\mathbf{x}_i^{s,t}, \mathbf{x}_i^{s,t}\right) = L_{D\text{-local}}\left(\mathbf{x}_i^{s,t}, d_i^{s,t}\right) + L_{D\text{-global}}\left(\mathbf{x}_i^{s,t}, d_i^{s,t}\right). \tag{13}$$

This step of complete loss is noted as

$$\min_{\theta_g, \theta_{f1}, \theta_{f2}} L_{\text{cls}}\left(\mathbf{x}_i^s, \mathbf{y}_i^s\right) + L_D\left(\mathbf{x}_i^{s,t}, \mathbf{x}_i^{s,t}\right). \tag{14}$$

In Step 2, to enable domain discriminators to more accurately distinguish between source and target domain samples, we fix the parameters of E, F_1, F_2, and train two domain discriminators. This is a part of domain adversarial learning, with the ultimate goal of ensuring that the feature vectors of the two domains have similar distributions to achieve global and local domain alignment. Specifically, optimizing the parameters of the domain discriminator θ_d is achieved by minimizing L_D

$$\min_{\theta_d} L_D\left(\mathbf{x}_i^{s,t}, d_i^{s,t}\right). \tag{15}$$

Note that θ_d here actually represents the parameters of both D_l and D_g modules. However, since D_l and D_g share the same structure and are always updated simultaneously, we simply use θ_d to represent the parameters of these two domain discriminators.

In Step 3, to optimize decision boundaries, we fix the parameters of E, D_g, D_l and train F_1, F_2. In particular, we use the CDD distance [46] to measure the classification difference of two classifiers, which expresses the differences as the class correlations of the different predictions and puts constraints on the discriminability of the target features. Compared to the L_1 distance, the CDD distance can ensure the discriminability of features and has better robustness to disturbances. It is defined as follows:

$$L_{\text{cdd}} = \sum_{m,n=1}^{k} \mathbf{A}_{mn} - \sum_{m=1}^{k} \mathbf{A}_{mn} = \sum_{m \neq n}^{k} \mathbf{A}_{mn} \tag{16}$$

where $\mathbf{A}$ is the predictive relevance matrix representing the differences in classifiers, the values of m and n range from 1 to k. $\mathbf{A}$ is defined as $\mathbf{A} = \mathbf{a}_1 \mathbf{a}_2^{\mathrm{T}}$, where $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^{k \times 1}$ represent the output probabilities of F_1 and F_2. $\mathbf{A}_{mn} = \mathbf{a}_1^m \mathbf{a}_2^n$ is the product of the probability of the sample being categorized as m by F_1 and the probability of the sample being categorized as n by F_2.

Specifically, this step maximizes the classification discrepancy between the two classifiers and the CST loss, while minimizing the prediction loss in the source domain. The complete loss is

$$\min_{\theta_{f1}, \theta_{f2}} L_{\text{cls}}\left(\mathbf{x}_i^s, \mathbf{y}_i^s\right) - \gamma L_{\text{dis}}\left(\mathbf{x}_i^t\right) - \beta L_{\text{Cycle}}\left(\mathbf{x}_i^s, y_i^s, \mathbf{x}_j^t, \hat{y}_j^t\right) \tag{17}$$

$$L_{\text{dis}}(\mathbf{x}_i^t) = \frac{1}{n_t} \sum_{i=1}^{n_t} L_{\text{cdd}}(\mathbf{x}_i^t). \tag{18}$$

After this training step, the decision boundary of the classifier moves away from the samples that do not have transferable features.

In Step 4, to achieve class-level adversarial alignment, we fix F_1, F_2, D_g, D_l, and update E to reduce the distribution difference between the source and target domains. Specifically, this step involves minimizing the CDD distance between the two classifiers, while minimizing the CST loss to promote the generation of more discriminative features. This step can be expressed as

$$\min_{\theta_g} \gamma L_{\text{dis}}\left(\mathbf{x}_i^t\right) + \beta L_{\text{Cycle}}\left(\mathbf{x}_i^s, y_i^s, \mathbf{x}_j^t, \widehat{y_j^t}\right). \tag{19}$$

After such a constraint, the features generated by E tend to be closer to the decision boundary, assisting the transfer of knowledge from the source domain to the target domain.

III. Experiments Result

A. Dataset

Two public datasets and two self-made datasets will be presented in this section.

1) Pavia Dataset: The Pavia dataset contains HSI data for two scenes, PU and PC, both acquired by the Reflective Optics Spectrographic Image System (ROSIS) sensor over the Pavia metropolitan area. We use PU as the source domain and PC as the target domain. The PU has a spatial resolution of 610 × 340 and consists of 103 spectral bands after removal of 12 noise bands. The PC has a spatial resolution of 1096 × 715 and consists of 102 spectral bands. In addition, seven common classes were selected for experimentation. Table I shows the names of the common classes in PU and PC along with the corresponding number of samples. The false-color

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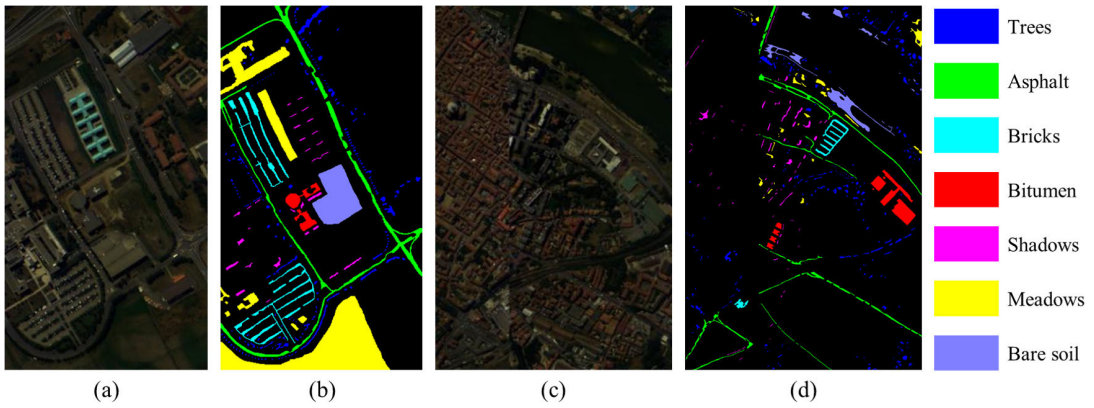


Fig. 4. Pavia dataset. (a) PU false-color image. (b) PU ground-truth map. (c) PC false-color image. (d) PC ground-truth map.

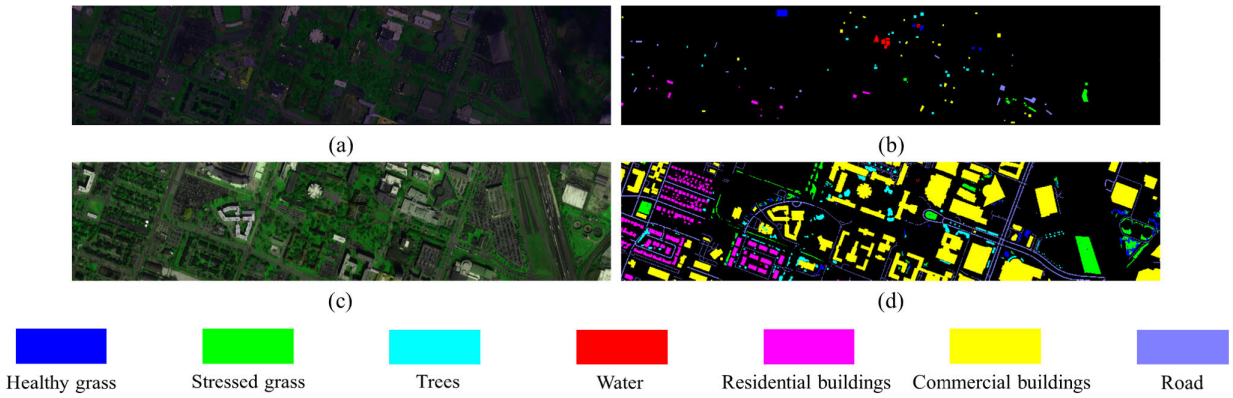


Fig. 5. Houston dataset. (a) Houston13 false-color image. (b) Houston13 ground-truth map. (c) Houston18 false-color image. (d) Houston18 ground-truth map.

TABLE I
NUMBER OF SAMPLES PER FEATURE CLASS IN THE PAVIA DATASET

ID	Class Name	PU	PC
1	Trees	3064	7598
2	Asphalt	6631	9248
3	Bricks	3682	2685
4	Bitumen	1330	7287
5	Shadows	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
	total	39332	39355

TABLE II
NUMBER OF SAMPLES PER FEATURE CLASS IN THE HOUSTON DATASET

ID	Class Name	Houston13	Houston18
1	Healthy grass	345	1353
2	Stressed grass	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Commercial buildings	408	32459
7	Road	443	6365
	total	2530	53200

composite images and the ground truth images are shown in Fig. 4.

2) Houston Dataset: This dataset consists of HSI from two scenes, Houston13 [47] and Houston18 [48]. Collected by the National Center for Airborne Laser Mapping (NCALM) and funded by the National Science Foundation, these datasets were acquired over the University of Houston campus and adjacent urban areas using different sensors in 2013 and 2018, respectively. We use Houston13 as the source domain and Houston18 as the target domain. After cropping and dimension adjustment, both Houston13 and Houston18 have a spatial resolution of 210 × 954 and contain 42 spectral bands. In our experiments, we took Houston13 as the source domain and Houston18 as the target domain, and selected seven classes common to both scenes for the experiments. Table II shows the names of the common categories and the number for each class. Their false-color and feature-true-value plots are shown in Fig. 5.

3) Willow Dataset: This dataset is our self-collected dataset of HSIs obtained from two natural willow forests in the Yellow River Delta Nature Reserve in Dongying City, Shandong Province, captured by a UAV carrying a Cubert S185 airborne imaging spectrometer on the same day in June 2023. This dataset contains HSI data for two scenes, Willow2 and Willow3. We use Willow2 as the source domain and Willow3 as the target domain. Willow2 has a spatial resolution of

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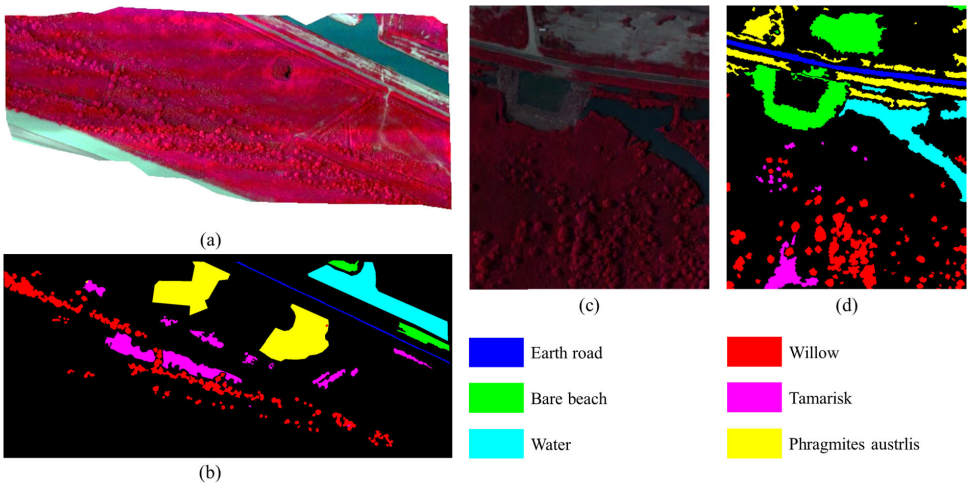


Fig. 6. Willow dataset. (a) Willow2 false-color image. (b) Willow2 ground-truth map. (c) Willow3 false-color image. (d) Willow3 ground-truth map.

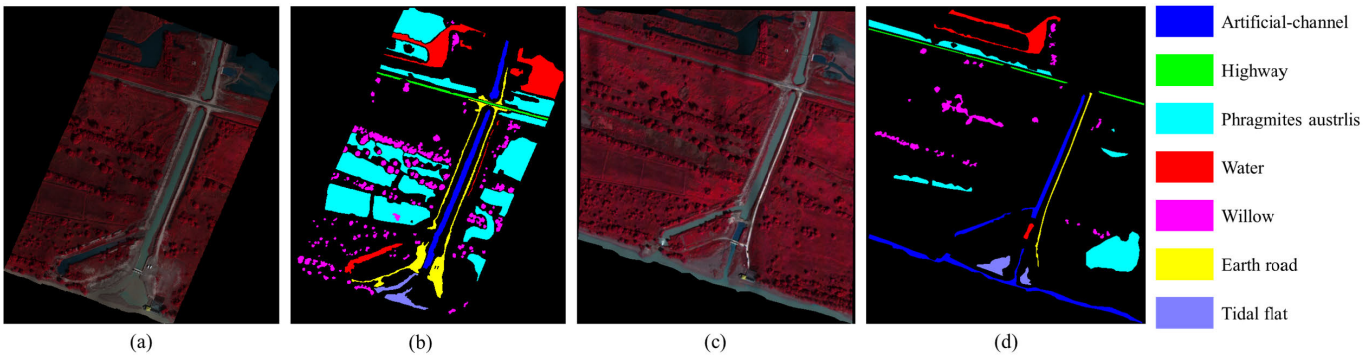


Fig. 7. YRW dataset. (a) YRW22 false-color image. (b) YRW22 ground-truth map. (c) YRW23 false-color image. (d) YRW23 ground-truth map.

TABLE III
NUMBER OF SAMPLES PER FEATURE CLASS IN THE WILLOW DATASET

ID	Class Name	Willow2	Willow3
1	Earth road	6798	4935
2	Bare beach	12767	21247
3	Water	57669	12683
4	Willow	55495	15904
5	Tamarisk	49129	4732
6	Phragmites austrlis	91216	14003
	total	273074	73504

TABLE IV
NUMBER OF SAMPLES PER FEATURE CLASS IN THE YRW DATASET

ID	Class Name	YRW22	YRW23
1	Artificial-channel	4003	7879
2	Highway	1114	1538
3	Phragmites austrlis	32896	15766
4	Water	9084	3478
5	Willow	8808	3638
6	Earth road	4982	9994
7	Tidal flat	1444	1634
	total	62331	43927

935 × 2060 and Willow3 has a spatial resolution of 586 × 495, with an original spectral dimension of 127 and 126 spectral bands remaining after removing one band that cannot be reflected by water, and six classes common to both scenes were selected. Table III shows the names of the common classes of Willow2 and Willow3 and the number of samples each contains. Their false-color and feature-true-value images are shown in Fig. 6.

4) YRW Dataset: This is our self-collected dataset comprising YRW22 and YRW23. It consists of HSIs captured in the same natural willow forest area in the Yellow River Delta Nature Reserve, Dongying City, Shandong Province, China, by unmanned aerial vehicles equipped with the Cubert S185 airborne imaging spectrometer in September 2022 and September 2023. We use YRW22 as the source domain and YRW23 as the target domain. YRW22 has a spatial resolution of 590 × 512, while YRW23 has a resolution of 637 × 560. After removing noise bands, both datasets retain 126 spectral bands. Seven common classes between the two scenes were selected for experimentation. Table IV shows the names of the common classes between YRW22 and YRW23, along with the corresponding sample counts. The false-color images and ground truth images are shown in Fig. 7.

B. Experimental Setup

All experiments in this article were performed on the device equipped with an NVIDIA GeForce RTX 3090 GPU. The

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TABLE V
CLASSIFICATION RESULT OBTAINED BY DIFFERENT METHODS ON PAVIA DATASET

class	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
1	85.08	89.50	90.10	91.85	**98.96**	98.03	92.73	97.92	95.28	96.93	94.37
2	77.64	91.89	84.70	95.62	93.45	95.95	85.07	96.53	92.80	93.65	**99.61**
3	29.05	82.12	67.23	78.66	80.07	79.59	89.91	79.18	83.35	99.66	**99.78**
4	63.57	63.78	76.55	68.83	85.32	76.67	83.49	**90.30**	85.11	84.70	84.81
5	90.78	**100.00**	99.97	**100.00**	**100.00**	**100.00**	98.57	**100.00**	99.97	99.90	**100.00**
6	51.59	72.39	68.71	84.27	62.17	85.70	75.47	88.03	90.10	**94.82**	94.43
7	77.49	65.17	79.31	79.65	88.58	81.99	88.09	**91.45**	90.55	79.58	85.19
OA(%)	72.04	80.15	81.99	85.53	89.30	88.81	87.32	89.49	91.14	91.17	**93.08**
AA(%)	67.88	80.69	80.94	85.56	86.93	88.27	87.62	89.06	91.02	92.75	**94.03**
K×100	66.31	76.19	78.34	82.59	87.10	86.55	84.79	87.32	89.37	89.52	**91.69**

TABLE VI
CLASSIFICATION RESULT OBTAINED BY DIFFERENT METHODS ON HOUSTON DATASET

class	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
1	50.18	64.30	65.85	75.24	48.19	72.51	77.83	**78.64**	62.01	93.05	62.75
2	58.39	44.27	90.30	75.04	90.84	79.48	61.25	56.91	82.16	48.24	**91.24**
3	66.96	**87.53**	65.58	78.38	68.47	67.21	81.78	52.60	54.09	60.05	63.96
4	81.82	90.91	63.64	81.82	40.91	59.09	**100.00**	27.27	45.45	50.00	68.18
5	95.40	46.08	69.27	80.08	96.67	94.63	57.60	**97.16**	92.18	93.23	94.71
6	52.96	62.33	57.83	52.36	50.26	53.42	**74.31**	58.99	58.74	**72.23**	68.92
7	38.05	45.77	41.07	53.01	65.03	68.03	50.70	**74.34**	**74.38**	61.13	57.86
OA(%)	56.61	58.43	60.57	59.25	61.49	62.91	69.09	64.98	65.96	70.67	**71.83**
AA(%)	63.39	63.03	64.79	70.85	66.05	70.62	71.92	64.13	67.00	68.28	**72.52**
K×100	40.35	40.77	45.30	44.70	48.27	49.43	53.67	51.02	52.28	56.03	**58.29**

training epoch was uniformly set to 100 and the batch size was 128. SGD was employed as the optimizer with a weight decay of 0.0005 and a momentum of 0.9. Additionally, for the Pavia Dataset, Willow Dataset, and YRW Dataset, 80 labeled samples per class were randomly selected from the source domain data for training with a learning rate of 0.05. For the Houston dataset, 180 samples were taken with a learning rate of 0.01. All target domain data were available for training but without any labels. The input patch size for all datasets is set to 7. More detailed parameter settings and comparisons are described in Section III-E.

C. Comparative Experiment

To assess the effectiveness of CSJA, we compared it with ten other methods, including nine cross-scene classification methods and one single-scene classification method. The patch size was uniformly set to 7×7 in the experiments. To assess the experimental results, we employed three common evaluation metrics: overall accuracy (OA), class-specific accuracy, and the Kappa coefficient.

In this article, seven cross-scenario methods, ADDA [49], DAN [28], DANN [31], MCD [33], TAADA [34], TSTNet [30], SCLUDA [42], CLDA [41], and MLUDA [50], are used to compare with CSJA to reflect the state-of-the-art of the methods in this article. ADDA and DANN both use adversarial learning strategies with domain discriminators, but they use different training approaches. DAN adds DA layers to deep networks and constrains the distributions of two domains using the MMD distance. MCD, TAADA, and CLDA are all designed based on the adversarial idea of maximizing classifier differences. Among them, CLDA designs a pseudo-label selection strategy based on confident learning. TSTNet uses the MMD distance and optimal transport to support the alignment of different domains. Both SCLUDA and MLUDA design class alignment strategies based on supervised comparison learning. Additionally, a method for single-domain classification method was used as a baseline for comparison. This method, referred to as "Source-only," involves training only on the source domain and directly classifying the target domain without using any domain alignment measures. In this study, the Source-only approach classifies the target domain using the Tri_CNN model [51].

The classification accuracies for OA, AA, Kappa, and each class for the four different datasets with different comparison methods are given in Tables V–VIII. Figs. 8–12 shows the classification results of the different methods on each dataset. Overall, CSJA outperforms the other methods on all four datasets. By analyzing the results of all the comparison experiments, we get the following discussion.

1) The Source-only method is less effective than the other methods. As can be seen from Table V, the OA of Tri_CNN in the Pavia dataset is 8.11% less accurate than the least effective cross-scene methods, which fully reflects the necessity and effectiveness of designing methods for cross-scene tasks. This proves that methods designed for cross-scene tasks take into account the distributional differences between the source and target domains, and that network structures specifically specified for the alignment and feature transfer

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TABLE VII
CLASSIFICATION RESULT OBTAINED BY DIFFERENT METHODS ON WILLOW DATASET

class	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
1	31.91	85.25	76.98	86.16	94.06	81.50	94.59	34.81	**96.62**	75.87	95.83
2	92.80	72.39	86.73	95.45	99.13	99.41	98.41	83.43	99.1	95.27	**99.62**
3	80.97	**100.00**	99.98	70.25	99.91	99.20	92.86	**100.00**	99.92	99.78	99.82
4	99.36	93.64	90.46	76.95	96.95	**98.82**	84.38	98.47	97.88	96.94	95.40
5	36.86	49.01	47.89	26.99	68.83	65.53	57.25	61.54	**77.90**	77.49	72.13
6	35.18	56.41	45.37	80.50	21.36	75.39	72.12	82.35	49.10	60.59	**93.24**
OA(%)	73.51	78.07	78.79	79.22	81.69	91.29	86.50	84.66	87.92	87.36	**95.50**
AA(%)	62.85	76.12	74.57	72.72	80.04	86.64	83.27	76.77	86.75	84.32	**92.67**
K×100	65.96	72.89	73.39	74.16	77.01	88.96	83.13	80.67	84.81	84.2	**94.33**

TABLE VIII
CLASSIFICATION RESULT OBTAINED BY DIFFERENT METHODS ON YRW DATASET

class	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
1	40.14	94.80	96.04	91.34	98.66	98.63	96.41	99.03	99.77	99.89	**99.99**
2	99.93	99.63	99.55	97.68	99.85	99.78	98.95	97.38	**100.00**	91.26	99.93
3	88.32	**83.12**	87.06	99.28	83.15	**100.00**	97.78	91.14	87.57	**100.00**	99.51
4	98.16	96.26	99.92	96.95	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
5	83.89	80.37	87.93	80.29	95.27	58.55	83.15	96.92	**97.80**	85.10	94.56
6	**100.00**	94.06	63.98	63.38	**100.00**	99.20	96.28	**100.00**	**100.00**	**100.00**	**100.00**
7	74.30	48.16	98.29	84.39	**100.00**	99.94	94.68	**100.00**	99.94	**100.00**	**100.00**
OA(%)	77.45	86.48	91.43	92.53	92.91	94.44	95.73	96.01	95.11	97.67	**99.13**
AA(%)	83.53	85.20	90.40	87.62	96.70	93.73	95.32	97.78	97.87	96.61	**99.14**
K×100	71.62	82.83	89.01	90.27	90.93	92.64	94.43	94.86	93.70	96.99	**98.87**

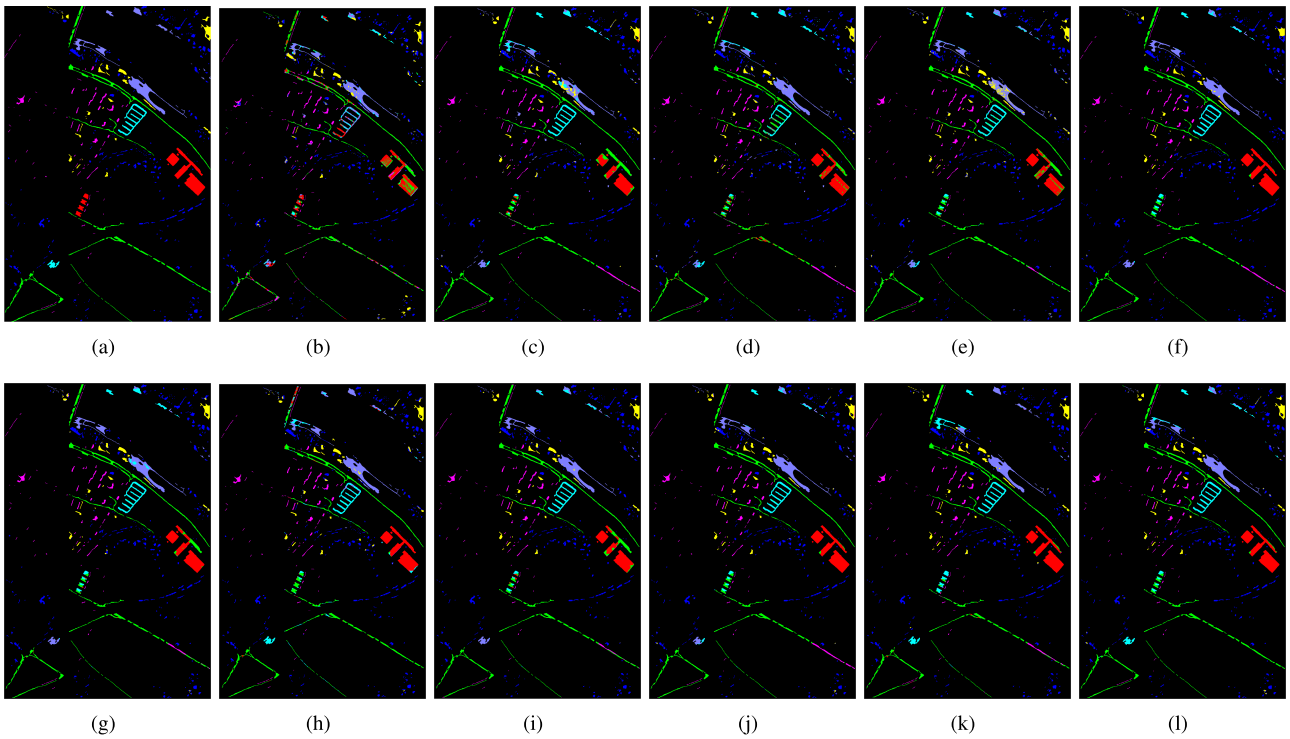
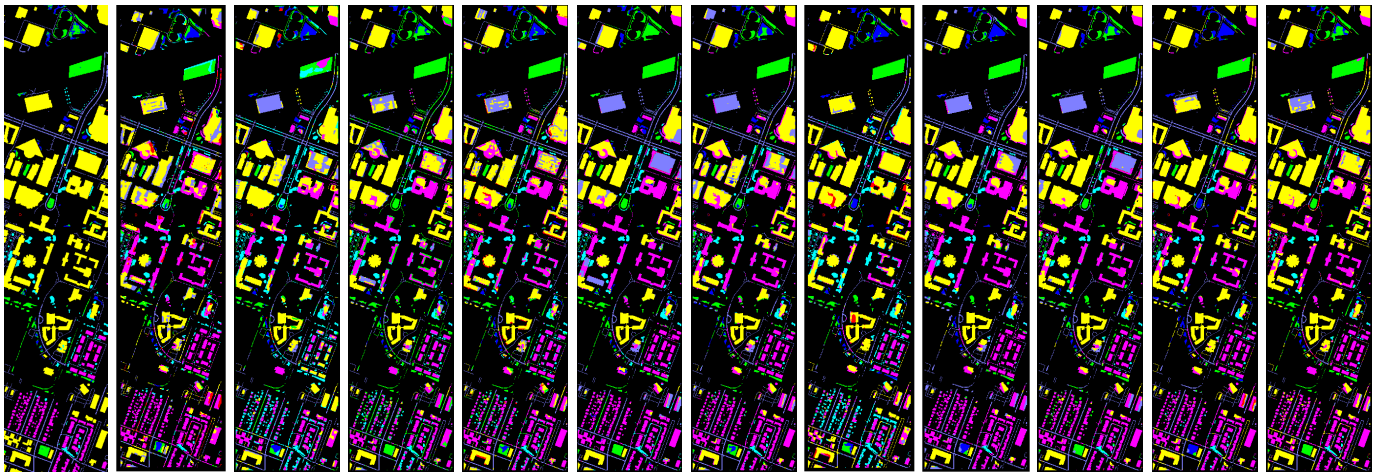


Fig. 8. Pavia data visualization and classification maps obtained by different models. (a) Ground truth image. (b) Tri_CNN. (c) ADDA. (d) DAN. (e) DANN. (f) MCD. (g) TAADA. (h) TSTnet. (i) SCLUDA. (j) CLDA. (k) MLUDA. (l) CSJA.

between the two distributions can improve classification performance.

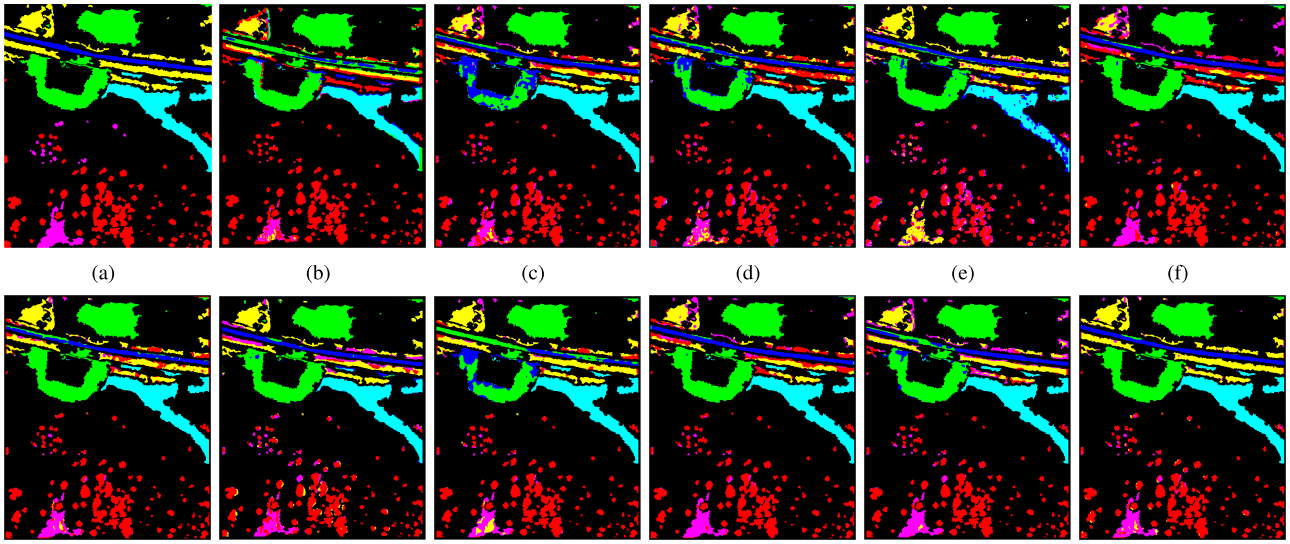
2) Methods using class alignment strategy work slightly better than methods using domain alignment strategy. ADDA, DAN, and DANN are methods using domain alignment strategy and MCD, TAADA, TSTNet, SCLUDA, CLDA, and MLUDA are methods using class alignment strategy. In the case of Pavia dataset,

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 9. Houston data visualization and classification maps obtained by different models. (a) Ground truth image. (b) Tri_CNN. (c) ADDA. (d) DAN. (e) DANN. (f) MCD. (g) TAADA. (h) TSTnet. (i) SCLUDA. (j) CLDA. (k) MLUDA. (l) CSJA.



(a) (b) (c) (d) (e) (f)

(g) (h) (i) (j) (k) (l)

Fig. 10. Willow data visualization and classification maps obtained by different models. (a) Ground truth image. (b) Tri_CNN. (c) ADDA. (d) DAN. (e) DANN. (f) MCD. (g) TAADA. (h) TSTnet. (i) SCLUDA. (j) CLDA. (k) MLUDA. (l) CSJA.

TABLE IX

SPACE COMPLEXITY AND FLOPS OF DIFFERENT METHODS ON PAVIA DATASET

Complexity	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
FLOPs	2.419G	7.108M	10.286M	276.583M	26.496M	27.966M	10.060M	80.328M	33.558M	161.660M	77.188M
Parameters	542.950M	214.679K	1.172M	11.139M	168.528K	214.087K	5.343M	379.274K	194.782K	732.794K	1.550M

for example, the OA value of the method with the worst results on this dataset among the class-aligned methods is 87.32%, which is 1.97% higher than the OA value of the method with the best performance of the domain-aligned methods on this dataset.

3) There are multiple vegetation classes in the self-made datasets Willow and YRW, e.g., Willow in the fourth class of the Willow dataset and the fifth class of the YRW dataset, Phragmites austrlis in the sixth class of the Willow dataset and the third class of the YRW dataset, and Tamarisk in the fifth class of the YRW dataset. The different vegetation conforms to the characteristics of plant spectral curves, and the undulating trends of their spectral curves are more similar, making them more confusing to classify. The classification results of these classes in Tables VII and VIII are worse compared to the other classes, and the poorly aligned methods even showed no distinction at all, which is consistent with the theory. On the other hand, MCD, TAADA, SCLUDA, CLDA, MLUDA, and CSJA are less affected than the

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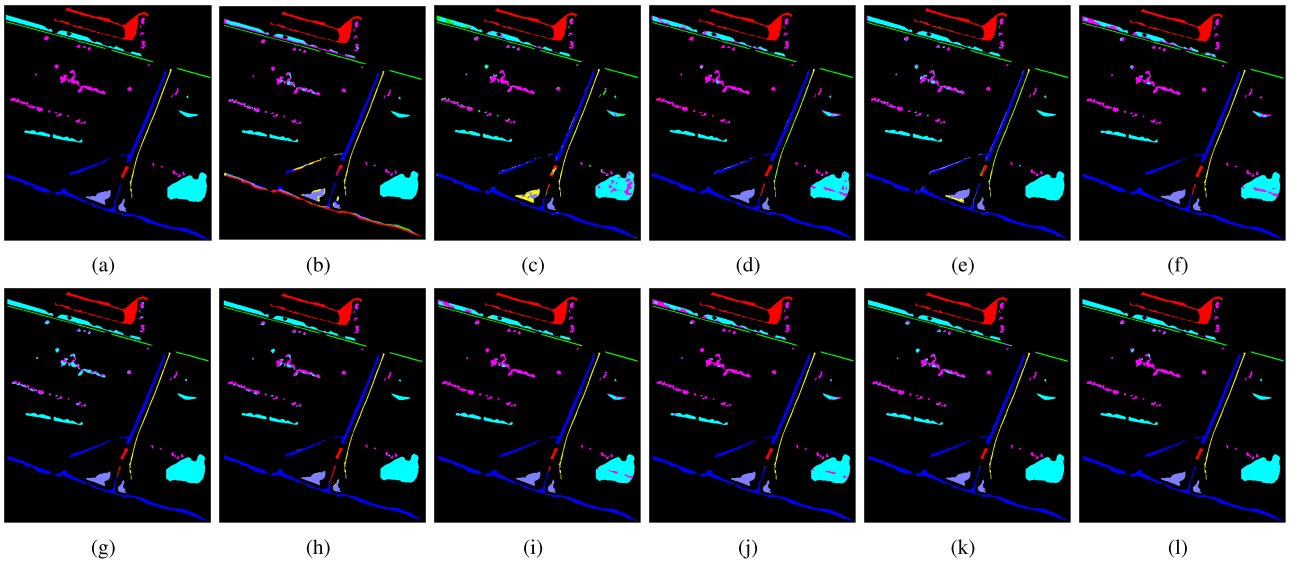


Fig. 11. YRW data visualization and classification maps obtained by different models. (a) Ground truth image. (b) Tri_CNN. (c) ADDA. (d) DAN. (e) DANN. (f) MCD. (g) TAADA. (h) TSTnet. (i) SCLUDA. (j) CLDA. (k) MLUDA. (l) CSJA.

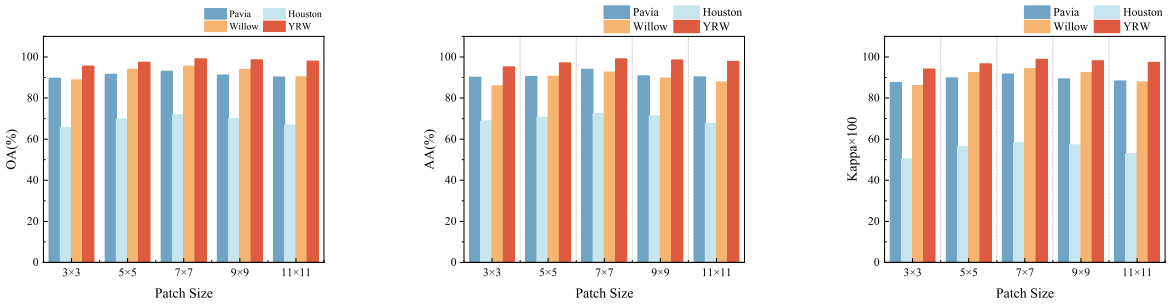


Fig. 12. Effect of different patch sizes on the classification performance of CSJA. (a) OA. (b) AA. (c) Kappa.

TABLE X
TRAINING AND TESTING TIME OVERHEAD OF DIFFERENT METHODS ON EACH DATASET

Dataset	Time	Tri_CNN	ADDA	DAN	DANN	MCD	TAADA	TSTNet	SCLUDA	CLDA	MLUDA	CSJA
Paiva	Training(s)	109.24	13.32	16.74	27.65	22.47	623.55	853.45	301.52	27.79	95.25	202.36
	Testting(s)	12.39	0.24	0.32	1.30	0.42	0.85	6.67	2.46	0.45	5.20	2.65
Houston	Training(s)	113.75	15.91	22.71	31.21	30.66	664.36	1147.22	657.82	31.99	86.01	318.49
	Testting(s)	7.70	0.41	0.45	2.82	0.50	0.90	4.78	2.22	1.20	3.26	3.64
Willow	Training(s)	102.05	8.54	13.81	21.51	17.98	576.44	568.04	337.15	23.71	97.46	143.01
	Testting(s)	28.90	0.28	0.75	4.02	0.82	2.43	4.03	5.39	0.82	11.20	4.27
YRW	Training(s)	135.66	10.46	11.79	31.61	19.55	449.65	530.56	371.18	22.23	110.46	179.63
	Testting(s)	11.47	0.37	0.59	1.61	0.32	0.74	4.95	5.15	0.33	4.42	1.70

other methods due to the extra attention paid to the alignment of the same category in different domains.

4) Among methods considering class alignment, our CSJA approach simultaneously considers both domain alignment and class alignment, outperforming the more effective CLDA method. This is shown in Table V for the Pavia dataset. The OA of CSJA is 1.94% higher than CLDA. In CLDA, class alignment operations are employed, and confident learning is used to select high-confidence target domain samples for pseudo-label training of the network. However, although pseudo-label selection based on confident learning is more accurate in sample selection than fixed-threshold methods, pseudo-labels may not be reliable in the case of spectral shifts in HSICC tasks, and even if scientifically selected, not all selected pseudo-labels may be conducive to classification. The CST strategy adopted in our method explicitly forces the generalization of pseudo-labels across scenes to achieve better classification performance.
5) In Tables IX and X, we compare the computational complexity, training time, and testing time of CSJA with other methods. The computational cost of a model can

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TABLE XI
EFFECT OF CONVOLUTIONALLY ENHANCED TRANSFORMER, GLOBAL–LOCAL DOMAIN DISCRIMINATOR AND CST ON OA (%)

Module	convolutionally enhanced transformer	global-local domain discriminator	cycle self-training	Pavia	Houston	Willow	YRW
Exp1	×	×	×	81.61	61.28	73.69	92.95
Exp2	✓	×	×	89.95	61.81	78.58	96.69
Exp3	✓	✓	×	90.36	66.78	83.88	97.22
Exp4	✓	×	✓	91.15	70.36	86.06	98.33
Exp5	✓	✓	✓	**93.08**	**71.83**	**95.50**	**99.13**

TABLE XII
ABLATION EXPERIMENT OF T-TRANSFORMER LAYER

	Pavia	Houston	Willow	YRW
without T-Transformer Layer	91.26	70.59	91.09	98.65
in the first layer	90.34	67.95	84.64	94.79
in the central layer	91.37	70.14	90.07	98.79
in the last layer	**93.08**	**71.83**	**95.50**	**99.13**

somewhat evaluate its performance. As seen in Table IX, using the Pavia dataset as an example, the FLOPs of our method are at a moderate level, lower than those of four other methods, including SCLUDA and MLUDA. However, the number of parameters is the fourth highest due to the use of a transformer architecture for the feature extractor, which contains a large number of parameters. Similarly, Table X shows that the training and testing times of CSJA are not the shortest, but the overall overhead is not excessive. Fortunately, although CSJA sacrifices some computational and spatial efficiency, it achieves the best classification performance.

D. Ablation Experiment

To further investigate the effectiveness of different modules in our method, we designed two sets of experiments. The five experiments in Table XI conduct ablation studies on three modules in the network: convolutional enhancement of the transformer, the global–local domain discriminator, and CST. The symbol "✓" indicates the use of the component in the experiment, while "×" indicates its absence. Exp1 is the baseline model without any enhancement modules, resulting in the worst performance among all experiments. The results of Exp2 show that the improved feature extractor can extract more discriminative features, which are conducive to domain alignment and lead to improved classification performance on the target domain. Exp3 and Exp4, respectively, investigate the effects of the global–local domain discriminator and the CST modules. The results show a significant improvement in performance after the inclusion of both modules, with the CST module particularly exhibiting better enhancement. Exp5 presents the classification performance when all modules are used. Overall, the experimental results indicate the effectiveness of different modules, and the combination of these modules can achieve substantial improvements in the overall classification performance.

Additionally, we discuss the necessity and optimal placement of the T-Transformer layer within the network, as shown in Table XII. As described in Section II-C, the T-Transformer layer is one of the layers in the feature extractor transformer network. It includes a local domain discriminator, which introduces transferable information captured by the local discriminator into the attention mechanism. We conducted experiments without the T-Transformer layer, as well as with it placed in the first, central, and last layers. This article uses a feature extractor transformer with seven layers, with the fourth layer being the central layer. The results show that placing the T-Transformer layer in the last layer of the network yields the best classification results, significantly outperforming the scenario without it, thereby underscoring the necessity of this module. However, when the T-Transformer layer is in the first layer, the classification performance is even worse than without it, indicating that it not only fails to function properly but also interferes with feature extraction. This might be due to the fact that, at the first layer, the features being processed are still shallow and highly susceptible to interference during domain discrimination. Moreover, placing the T-Transformer layer in the central layer does not provide significant advantages. Therefore, we ultimately chose to place the T-Transformer layer in the last layer to achieve the best network performance.

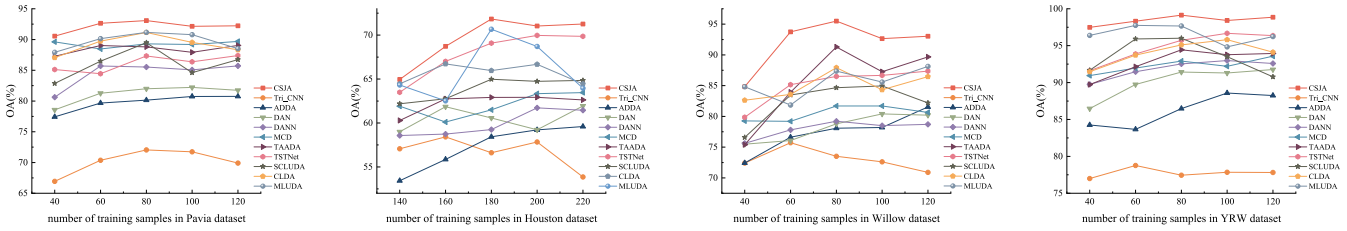
E. Parameter Analysis

This section explores the effects of four variables on the performance of the CSJA algorithm: patch size, number of training samples, learning rate, the three tradeoff parameters γ, α, β in (11), and the number of layers of the network.

For patch size, we set up five different sizes of patches for the experiments, which are 3×3, 5×5, 7×7, 9×9, and 11×11. The variation of classification results with patch size on the four datasets is shown in Fig. 12. The results show that the trend of classification accuracy with patch size is basically increasing first and then decreasing, and the best effect is achieved at patch size 7×7. The experimental results show that the patch size has a significant impact on classification accuracy. When the patch size is too small, it contains insufficient information, which prevents the network from effectively using spatial information for classification. Conversely, when the patch size is too large, it contains overly complex and redundant information, which not only affects the network's decision-making process, but also increases computational complexity. Therefore, choosing the right patch size enhances classification accuracy.

For the number of training samples in the source domain, we analyzed five different sample sizes for each dataset. The effect on classification is shown in Fig. 13. For the Pavia,

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(a) (b) (c) (d)

Fig. 13. Classification accuracy of different methods with different training samples on each dataset. (a) Pavia dataset. (b) Houston dataset. (c) Willow dataset. (d) YRW dataset.

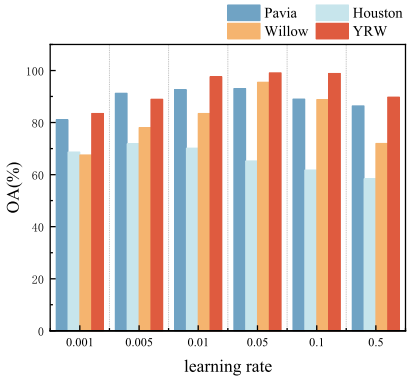


Fig. 14. Effect of learning rate on the Classification Effectiveness of CSJA.

Willow, and YRW datasets, we compared the classification results when extracting 40, 60, 80, 100, and 120 samples from each class. For the Houston dataset, we compared the classification results when extracting 140, 160, 180, 200, and 220 samples from each class. The classification results of all methods exhibit a similar trend: as the number of training samples in the source domain increases, classification accuracy improves, peaks, and then fluctuates around the peak. However, Tri_CNN deviates from this pattern. This is because Tri_CNN is a "Source-only" method with no measures to mitigate domain shift, so increasing the number of training samples does not significantly improve classification performance in the target domain. For CSJA, the Pavia, Willow, and YRW datasets all optimize at a sample size of 80, and the Houston dataset optimizes at 180. Additionally, when the number of training samples is excessive, some methods, such as SCLUDA, CLDA, and MLUDA, exhibit a decline in accuracy. This could be because these methods involve pseudo-labels, and as the number of training samples increases, the likelihood of learning incorrect information from pseudo-labels also rises. This erroneous information may lead to a decrease in the model's accuracy when generalizing to target domain data. In all sample size scenarios, CSJA shows a clear advantage, with its effectiveness being most pronounced when the sample size is between 60 and 100. This indicates that with a limited sample size, CSJA can effectively extract useful features from the source domain for classification.

For the learning rate, we fixed other parameters and set the learning rate to 0.001, 0.005, 0.01, 0.05, and 0.1. The trend of the classification results with respect to the learning rate is shown in Fig. 14. The learning rate significantly impacts model accuracy as it governs the step size of gradient descent during training. An excessively large learning rate may induce model divergence or oscillation near the optimal solution, whereas an excessively small learning rate, although conducive to convergence, could decelerate training speed and potentially trap the model in a local optimum. The Pavia, Willow, and YRW datasets were all optimized at a learning rate of 0.05, and the Houston dataset was optimized at a learning rate of 0.005.

TABLE XIII

CLASSIFICATION RESULTS FOR DIFFERENT VALUES OF β

β	0.1	0.3	0.5	0.7	0.9
Pavia	91.99	92.5	**93.08**	92.28	90.82
Houston	69.93	71.30	**71.83**	70.07	68.51
Willow	91.77	92.91	**95.50**	95.10	92.75
YRW	98.54	**99.13**	99.09	98.78	98.58

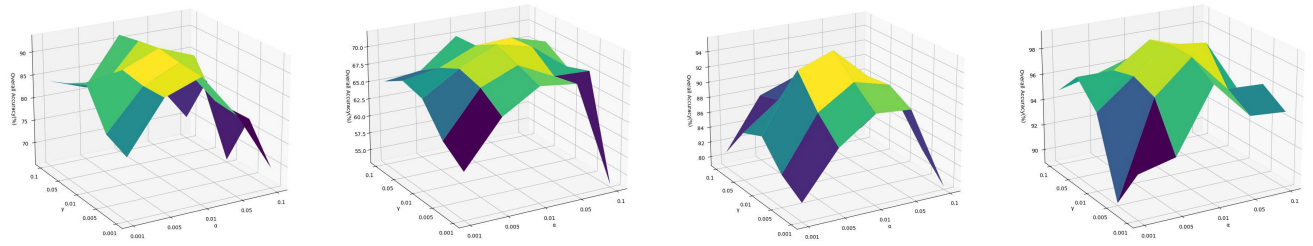
TABLE XIV

EFFECT OF THE NUMBER OF NETWORK LAYERS ON THE PAVIA DATASET ON THE RESULTS

T-Transformer Layer	Transformer Layer				
	2	4	6	8	10
1	90.41	91.53	**93.08**	92.62	90.46
2	90.2	90.72	92.01	92.94	91.36
3	90.25	91.13	92.06	92.5	89.86

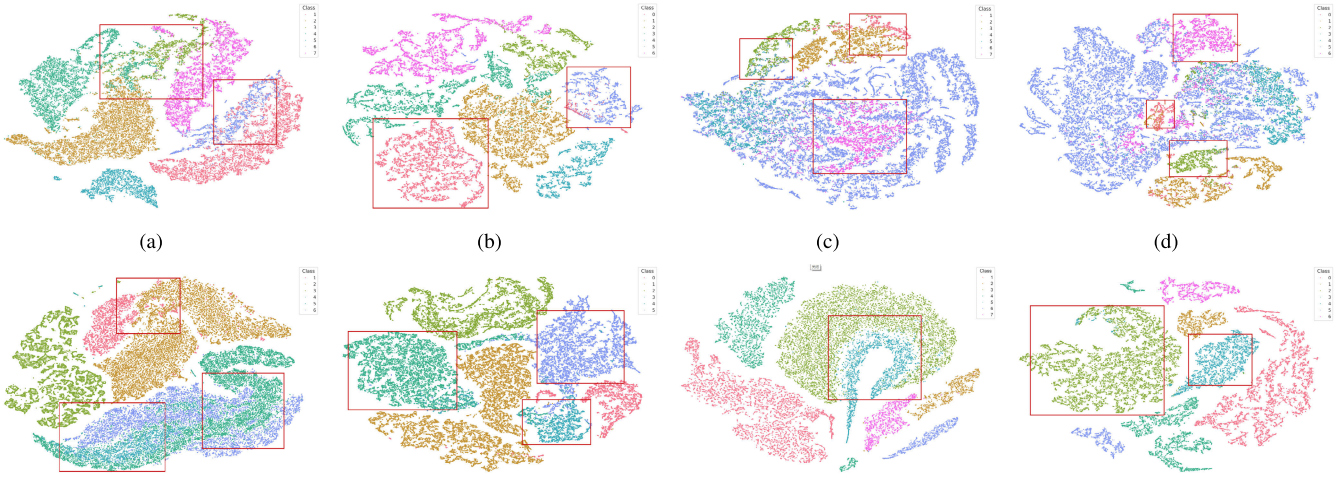
For γ, α, and β, they control the class adversarial loss, domain adversarial loss, and CST loss, respectively. To determine the optimal values for these parameters, we first fixed β at 0.5 and then examined the sensitivity of γ and α. Five values ranging from 0.001 to 0.1 were chosen for both γ and α, and the specific values and experimental results are shown in Fig. 15. The results indicate that the relationship between γ, α, and the OA approximates a convex function across all four datasets. When γ and α are within the range of 0.005 to 0.05, OA remains relatively stable and close to the optimal values. However, once beyond this range, OA drops sharply. Based on these results, the optimal values for both γ and α across the four datasets are 0.01. Next, we fixed γ and α and conducted sensitivity tests for β. The experimental results for different values of β across the four datasets are shown in Table XIII. The results suggest that the value of β also affects classification results, but good classification performance is maintained for values ranging from 0.1 to 0.9. Specifically, the

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(a) (b) (c) (d)

Fig. 15. Effect of different values of γ and α on overall classification accuracy. (a) Pavia dataset. (b) Houston dataset. (c) Willow dataset. (d) YRW dataset.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 16. Feature visualization results before and after alignment on different datasets based on t_SNE. (a) Pavia dataset before alignment. (b) Pavia dataset after alignment. (c) Houston dataset before alignment. (d) Houston dataset after alignment. (e) Willow dataset before alignment. (f) Willow dataset after alignment. (g) YRW dataset before alignment. (h) YRW dataset after alignment.

optimal value of β is 0.5 for the Pavia, Houston, and Willow datasets, and 0.3 for the YRW dataset.

For the number of layers in the network, we set the Transformer layer and T-Transformer layer counts separately. We experimented with the number of network layers using the Pavia dataset as an example. The experimental results in Table XIV show that the model achieves optimal performance with 6 Transformer layers and 1 T-Transformer layer, resulting in a total of seven layers. In Transformer networks, having too few layers limits the model's feature extraction capabilities, while too many layers can lead to overfitting and increased computational costs. Thus, six layers strike the best balance in our model. Additionally, the T-Transformer layer effectively achieves domain alignment through adversarial training. Setting it to 1 layer already yields good results, and increasing the layer count further does not significantly improve performance but does add computational overhead.

F. Feature Visualization

Visualizing features through t-SNE provides a straightforward way to visually demonstrate the effect of feature processing. Fig. 16 illustrates the feature visualization results before and after alignment for four datasets. From the feature distribution before alignment, it can be observed that there are noticeable confusions between some classes, such as classes four, five, and six of the YRW dataset being nearly completely mixed together in Fig. 16(e). After applying the alignment strategy proposed in CSJA, as shown in Fig. 16(f), the features of these classes are clearly separated, and the boundaries between classes become clearer. This improvement is primarily attributed to CSJA's combined alignment strategy of class alignment and domain alignment, which enhances the class separability of target domain features.

IV. Conclusion

In this article, we propose a DA architecture, CSJA, based on CST with joint adversarial, to address the spectral shift issue across different scenes in HSICC tasks. CSJA employs an adversarial learning strategy to minimize the discrepancy between two distributions, using global and local domain discriminators against the feature extractor, and adversarial training between the dual classifiers and the feature extractor to achieve domain and class alignment. In addition, an enhanced self-training method is used to gradually refine the pseudo-labels and force their generalization to learn more domain-invariant knowledge. Extensive experiments and analysis show that the CSJA method outperforms other state-of-the-art HSICC methods on both public and self-made datasets.

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Zhongwei Li received the Ph.D. degree from China University of Petroleum (East China), Qingdao, China, in 2011.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China). His research interests include remote sensing image processing and ocean numerical forecasting and cloud computing.



Yajie Yang received the B.E. degree in software engineering from China University of Petroleum (East China), Qingdao, China, in 2022, where she is currently pursuing the master's degree with the College of Computer Science and Technology.

Her research interests include the hyperspectral image cross-scene classification.



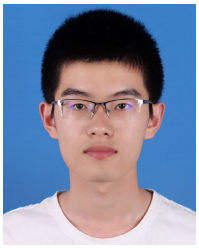
Leiquan Wang (Member, IEEE) received the Ph.D. degree in communication and electrical systems from Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2016.

He is currently a Lecturer with the College of Computer Science and Technology, China University of Petroleum (East China), Qingdao, China. His research interests include multimodal fusion, cross-modal retrieval, and image/video caption.



Mingming Xu (Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

She is currently an Associate Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. Her research interests include hyperspectral image processing and wetland remote sensing.



Ziqi Xin received the B.E. degree in communication engineering from China University of Petroleum (East China), Qingdao, China, in 2020, where he is currently pursuing the Ph.D. degree with the College of Oceanography and Space Informatics.

His research interests include hyperspectral images classification.



Jie Wei received the B.E. degree in computer science and technology from Shandong University of Finance and Economics, Jinan, China, in 2022. She is currently pursuing the master's degree with the College of Computer Science and Technology, China University of Petroleum (East China), Qingdao, China.

Her research interests include the hyperspectral images classification.



Yuewen Wang received the B.E. degree in Internet of Things engineering from China University of Petroleum (East China), Qingdao, China, in 2022, where she is currently pursuing the master's degree with the College of Computer Science and Technology.

Her research interests include the hyperspectral and LiDAR data classification.

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